

glmCH1

Load and clean data

```
#load packages  
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
## v dplyr      1.1.4      v readr      2.1.5  
## v forcats    1.0.0      v stringr   1.5.1  
## v ggplot2    3.5.2      v tibble    3.3.0  
## v lubridate  1.9.4      v tidyr     1.3.1  
## v purrr      1.0.4  
## -- Conflicts ----- tidyverse_conflicts() --  
## x dplyr::filter() masks stats::filter()  
## x dplyr::lag()     masks stats::lag()  
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)  
library(DHARMA)
```

```
## This is DHARMA 0.4.7. For overview type '?DHARMA'. For recent changes, type news(package = 'DHARMA')
```

```
library(glmmTMB)  
  
#read in data  
data <- read.csv("06_cleandata/combined_plot_data.csv")  
  
#check that variables are in appropriate format  
str(data)
```

```
## 'data.frame':   210 obs. of  34 variables:  
## $ site_id      : chr  "bates_57b" "bates_57b" "bates_57b" "bates_57b" ...  
## $ round        : chr  "2024_r1" "2024_r1" "2024_r1" "2024_r2" ...  
## $ plot_id      : chr  "p1" "p2" "d" "p1" ...  
## $ bee_date     : chr  "2024-04-15" "2024-04-15" "2024-04-15" "2024-06-06" ...  
## $ julian.date   : int   106 106 106 158 158 158 173 173 173 101 ...  
## $ bee_season_type : chr  "nestsearching" "nestsearching" "nestsearching" "worker" ...  
## $ area         : chr  "delta_1" "delta_1" "delta_1" "delta_1" ...  
## $ cover_type    : chr  "grass" "grass" "grass" "grass" ...  
## $ bare_perc     : num   2 1.42 0 11.92 20.5 ...  
## $ grass_perc    : num  46.8 52.2 90.4 52.7 56.8 ...  
## $ forb_perc     : num   9.33 7.92 8 16.75 8.12 ...  
## $ grass_forb_perc : num  56.2 60.2 98.4 69.4 64.9 ...
```

```
## $ grass_h : num 17.9 19.5 28.8 25.3 25.7 ...
## $ forb_h : num 9.83 7.42 9.58 26.17 17.42 ...
## $ floral_perc : num 0.375 1.0417 0.125 0.0417 0 ...
## $ bombus_floral_perc : num 0.375 1.0417 0.0833 0.0417 0 ...
## $ grass_forb_floral_perc : num 56.5 61.2 98.5 69.5 64.9 ...
## $ flood_perc : num 0 0 0 0 0 0 0 0 0 ...
## $ dry_root_perc : num 25.42 37.17 7.08 16.25 5.42 ...
## $ bryo_perc : num 0 0 0 0 0 0 0 0 0 ...
## $ bare_flood_dry_bryo_perc : num 27.42 38.58 7.08 28.17 25.92 ...
## $ bare_dry_bryo_perc : num 27.42 38.58 7.08 28.17 25.92 ...
## $ floral_richness : int 1 1 3 2 3 14 2 2 12 2 ...
## $ bombus_floral_richness : int 1 1 2 2 3 11 2 2 10 2 ...
## $ floral_abun : num 3 3 3.32 2.01 2.01 ...
## $ bombus_floral_abun : num 3 3 3.04 2.01 2.01 ...
## $ nest_searching_queens : int 3 0 4 2 0 0 0 0 0 ...
## $ workers : int 1 0 0 0 1 2 0 0 0 ...
## $ forage_vaco : int 1 2 0 2 4 0 0 0 0 3 ...
## $ num_burrows : int 41 120 93 41 120 93 41 120 93 41 ...
## $ num_inactive_burrows : int 10 64 0 10 64 0 10 64 0 10 ...
## $ num_active_dm : int 5 6 0 5 6 0 5 6 0 5 ...
## $ num_dm_size : int 3 3 0 3 3 0 3 3 0 3 ...
## $ num_inactive_dm : int 1 2 0 1 2 0 1 2 0 1 ...
```

```
#change character columns to factors
surveys <- data %>%
  mutate(
    site_id = as.factor(site_id),
    round = as.factor(round),
    plot_id = as.factor(plot_id),
    bee_date = as.Date(bee_date),
    bee_season_type = as.factor(bee_season_type),
    area = as.factor(area))
```

Scale continuous predictors

```
surveys$julian.date_scaled <- as.numeric(scale(surveys$julian.date))
surveys$bare_perc_scaled <- as.numeric(scale(surveys$bare_perc))
surveys$grass_perc_scaled <- as.numeric(scale(surveys$grass_perc))
surveys$forb_perc_scaled <- as.numeric(scale(surveys$forb_perc))
surveys$grass_forb_perc_scaled <- as.numeric(scale(surveys$grass_forb_perc))
surveys$grass_h_scaled <- as.numeric(scale(surveys$grass_h))
surveys$forb_h_scaled <- as.numeric(scale(surveys$forb_h))
surveys$floral_perc_scaled <- as.numeric(scale(surveys$floral_perc))
surveys$grass_forb_floral_perc_scaled <- as.numeric(scale(surveys$grass_forb_floral_perc))
surveys$bombus_floral_perc_scaled <- as.numeric(scale(surveys$bombus_floral_perc))
surveys$flood_perc_scaled <- as.numeric(scale(surveys$flood_perc))
surveys$dry_root_perc_scaled <- as.numeric(scale(surveys$dry_root_perc))
surveys$bryo_perc_scaled <- as.numeric(scale(surveys$bryo_perc))
surveys$bare_flood_dry_bryo_perc_scaled <- as.numeric(scale(surveys$bare_flood_dry_bryo_perc))
surveys$floral_richness_scaled <- as.numeric(scale(surveys$floral_richness))
surveys$bombus_floral_richness_scaled <- as.numeric(scale(surveys$bombus_floral_richness))
surveys$floral_abun_scaled <- as.numeric(scale(surveys$floral_abun))
surveys$bombus_floral_abun_scaled <- as.numeric(scale(surveys$bombus_floral_abun))
surveys$nest_searching_queens_scaled <- as.numeric(scale(surveys$nest_searching_queens))
```

```

surveys$workers_scaled <- as.numeric(scale(surveys$workers))
surveys$forage_vaco_scaled <- as.numeric(scale(surveys$forage_vaco))
surveys$num_burrows_scaled <- as.numeric(scale(surveys$num_burrows))
surveys$num_inactive_burrows_scaled <- as.numeric(scale(surveys$num_inactive_burrows))
surveys$num_active_dm_scaled <- as.numeric(scale(surveys$num_active_dm))
surveys$num_dm_size_scaled <- as.numeric(scale(surveys$num_dm_size))
surveys$num_inactive_dm_scaled <- as.numeric(scale(surveys$num_inactive_dm))

```

Filter to only include field data

```

field_data <- surveys %>%
  filter(plot_id != "d")

```

GLM #1: Number of nest-searching queens

Plotting a zero-inflated poisson model

```

nsq_lm_zeroPois <- glmmTMB(nest_searching_queens ~ grass_forb_perc_scaled +
  bombus_floral_abun_scaled +
  num_burrows_scaled +
  num_inactive_burrows_scaled +
  bee_season_type,
  ziformula = ~1, # zero-inflation model
  family = poisson, #poisson
  data = field_data)

summary(nsq_lm_zeroPois)

## Family: poisson ( log )
## Formula:
## nest_searching_queens ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
##      num_burrows_scaled + num_inactive_burrows_scaled + bee_season_type
## Zero inflation: ~1
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    151.3    171.9    -68.7    137.3      133
##
##
## Conditional model:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      0.2772   0.4464   0.621  0.53462
## grass_forb_perc_scaled      0.2930   0.2493   1.175  0.23980
## bombus_floral_abun_scaled      0.2277   0.2092   1.088  0.27652
## num_burrows_scaled      2.3470   1.1074   2.119  0.03405 *
## num_inactive_burrows_scaled    -1.9770   0.9995  -1.978  0.04793 *
## bee_season_typerworker     -2.2642   0.6247  -3.624  0.00029 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:

```

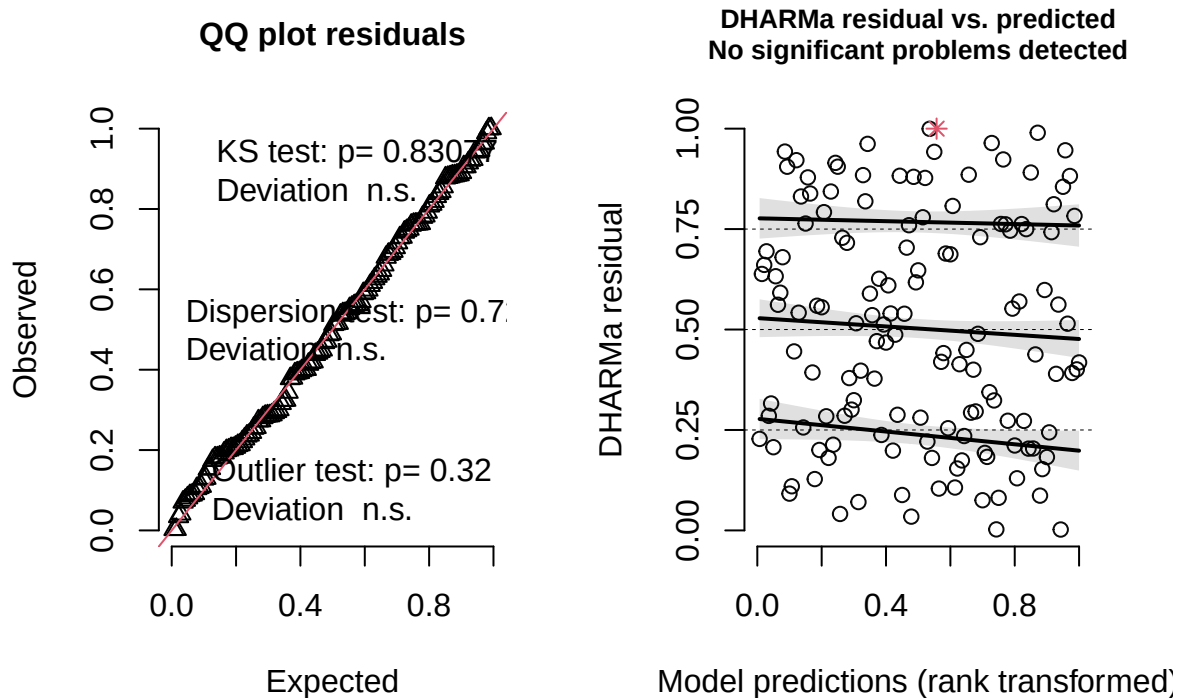
```
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.5993      0.8062  -0.743   0.457
```

Diagnostics

```
#simulate residuals
```

```
residuals_nsq_zeroPois <- simulateResiduals(fittedModel = nsq_lm_zeroPois, plot = TRUE)
```

DHARMA residual



```
##no evidence of overdispersion or significant outliers
```

Plotting a binary model to predict if $\#NSQ > 0$

```
#create binary variable for nsq presence
```

```
field_data$nsq_binary <- ifelse(field_data$nest_searching_queens > 0, 1, 0)
```

```
nsq_lm_binary <- glmmTMB(nsq_binary ~ grass_forb_perc_scaled +
  bombus_floral_abun_scaled +
  num_burrows_scaled +
  num_inactive_burrows_scaled +
  bee_season_type,
  family = binomial,
  data = field_data)
```

```
summary(nsq_lm_binary)
```

```
## Family: binomial ( logit )
## Formula:
## nsq_binary ~ grass_forb_perc_scaled + bombus_floral_abun_scaled +
##   num_burrows_scaled + num_inactive_burrows_scaled + bee_season_type
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##      97.8     115.4     -42.9     85.8      134
##
##
## Conditional model:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -0.3569    0.6149  -0.580 0.561590
## grass_forb_perc_scaled    1.0014    0.3596   2.785 0.005357 **
## bombus_floral_abun_scaled    0.4924    0.2880   1.710 0.087303 .
## num_burrows_scaled    1.3950    1.7315   0.806 0.420430
## num_inactive_burrows_scaled -1.6596    1.4770  -1.124 0.261167
## bee_season_typeworker    -2.6963    0.7890  -3.417 0.000633 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

GLM #2: Number of workers

Model selection between poisson, negative binomial and zero-inflated models.

First, plot it as a poisson model.

```
worker_lm_pois <- glmmTMB(workers ~ nest_searching_queens_scaled +
  bombus_floral_abun_scaled +
  num_burrows_scaled +
  num_inactive_burrows_scaled +
  bee_season_type,
  family = poisson,
  data = field_data
)

summary(worker_lm_pois)
```

```
## Family: poisson ( log )
## Formula:
## workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled +
##   num_burrows_scaled + num_inactive_burrows_scaled + bee_season_type
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##      200.5     218.2     -94.3     188.5      134
##
##
## Conditional model:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      -2.3978    0.5046  -4.752 2.02e-06 ***
## nest_searching_queens_scaled -0.1122    0.4338  -0.259 0.795887
```

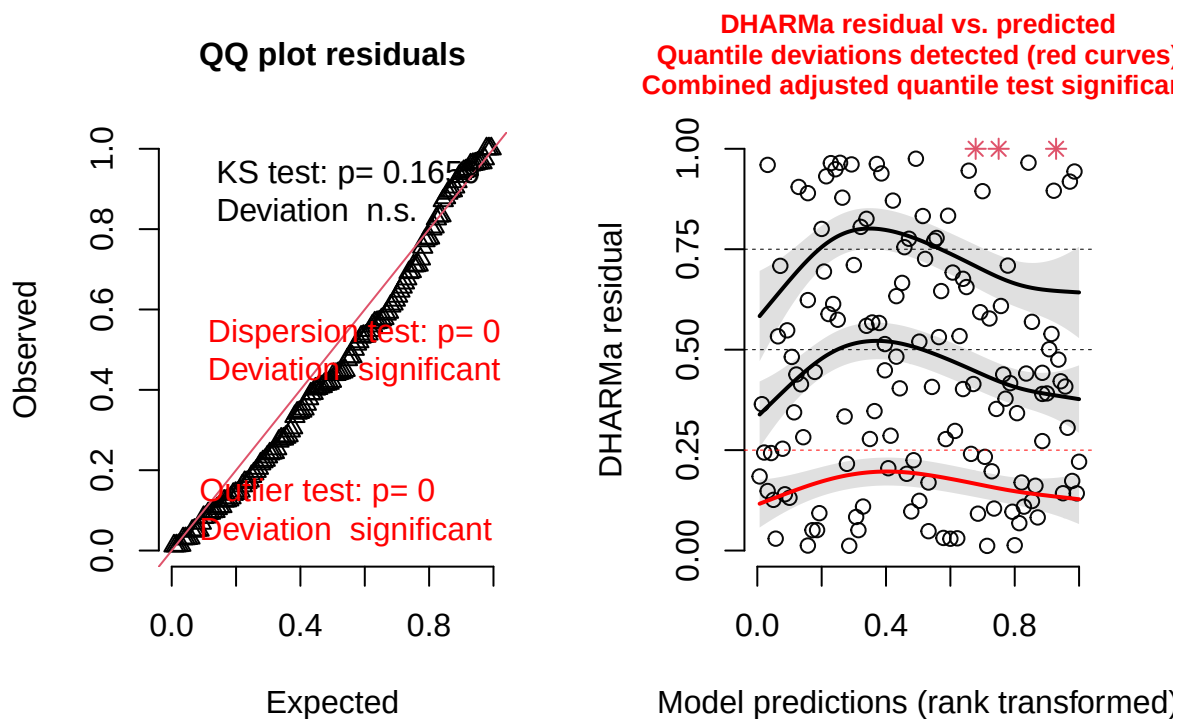
```
## bombus_floral_abun_scaled      0.4441      0.1938      2.291 0.021954 *
## num_burrows_scaled            0.6413      0.8349      0.768 0.442429
## num_inactive_burrows_scaled   -0.3711      0.6694     -0.554 0.579283
## bee_season_typerworker         1.5872      0.4811      3.299 0.000971 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Run diagnostic tests

```
#simulate residuals
```

```
residuals_workers_pois <- simulateResiduals(fittedModel = worker_lm_pois, plot = TRUE)
```

DHARMA residual



```
##residual plots show that there is significant overdispersion --> i should plot the negative binomial
```

Plot negative binomial

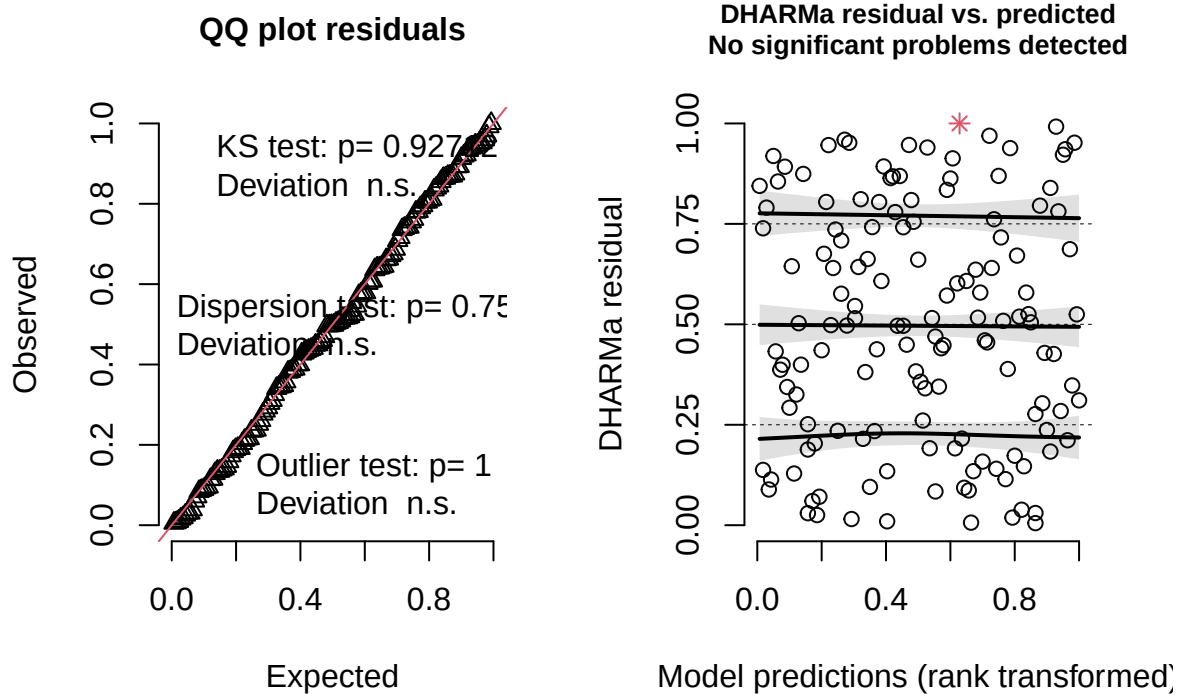
```
worker_lm_negbin <- glmmTMB(workers ~ nest_searching_queens_scaled +
  bombus_floral_abun_scaled +
  num_burrows_scaled +
  num_inactive_burrows_scaled +
  bee_season_type,
  family = nbinom1,
  data = field_data
)
```

Diagnostics

```
#simulate residuals
```

```
residuals_workers_negbin <- simulateResiduals(fittedModel = worker_lm_negbin, plot = TRUE)
```

DHARMA residual

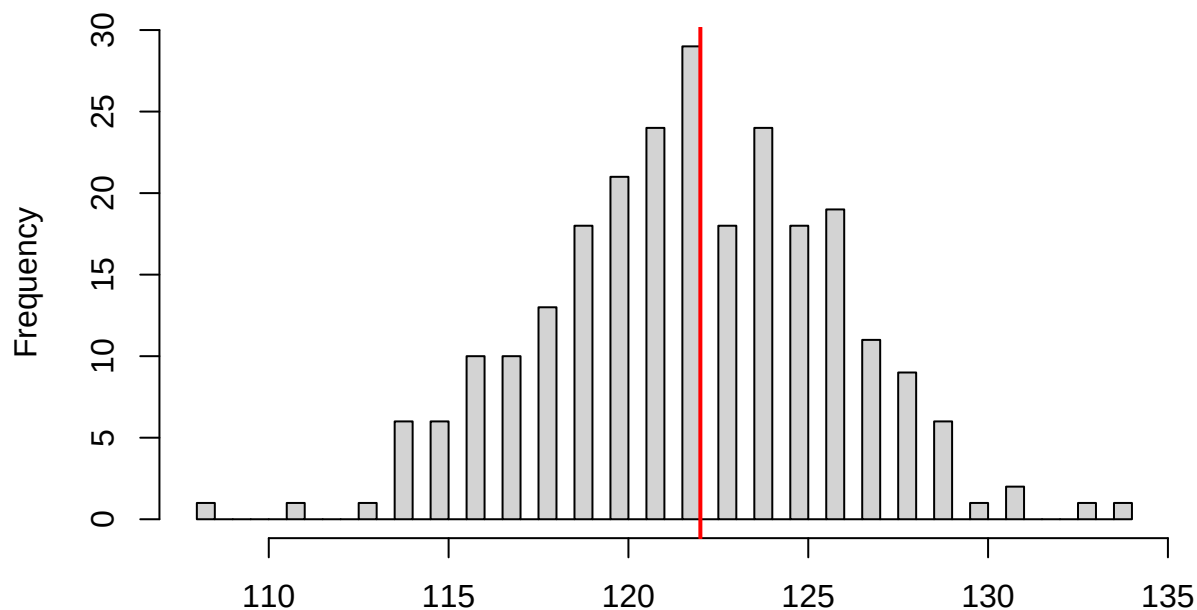


```
##residual plots show that there is no overdispersion and no excessive outliers. this shows that the ne
```

Test zero-inflation

```
testZeroInflation(residuals_workers_negbin)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 1

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 1.0003, p-value = 1
## alternative hypothesis: two.sided
```

##p-value = 1. No evidence of zero-inflation. Do not fit zero-inflated model, leave as negative binomial

Look at results of final model

```
summary(worker_lm_negbin)
```

```
## Family: nbinom1 ( log )
## Formula:
## workers ~ nest_searching_queens_scaled + bombus_floral_abun_scaled +
## num_burrows_scaled + num_inactive_burrows_scaled + bee_season_type
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    163.0    183.6    -74.5    149.0      133
##
##
```



```
## Dispersion parameter for nbinom1 family (): 1.94
##
## Conditional model:
##
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -2.04281    0.61003  -3.349 0.000812 ***
## nest_searching_queens_scaled  0.05988    0.44176   0.136 0.892178
## bombus_floral_abun_scaled    0.49887    0.27228   1.832 0.066917 .
## num_burrows_scaled          0.09062    1.08744   0.083 0.933589
## num_inactive_burrows_scaled  0.09220    0.81302   0.113 0.909706
## bee_season_typeworker       0.92043    0.54647   1.684 0.092118 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

GLM #3: Number of burrows

First, plot poisson

```
burrows_lm_pois <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
  julian.date_scaled,
  family = poisson, #poisson
  data = field_data)
```

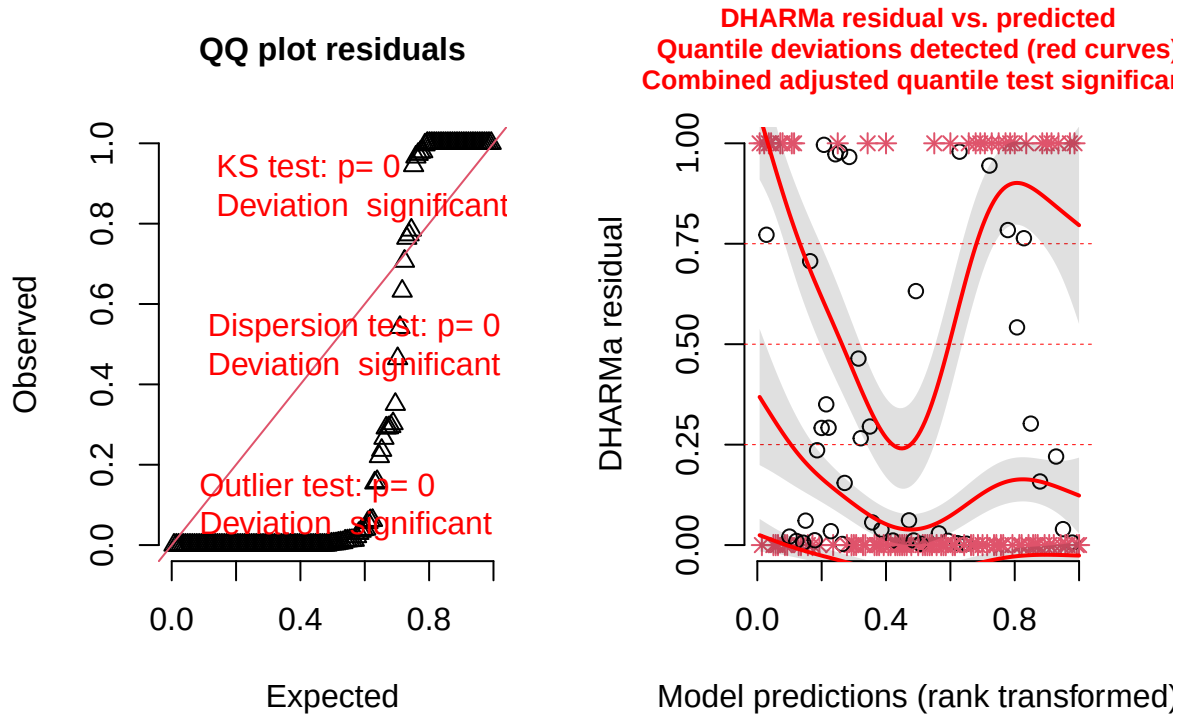
Diagnostics

```
#simulate residuals

residuals_burrows_pois <- simulateResiduals(fittedModel = burrows_lm_pois, plot = TRUE)

## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Fitting terminated with step failure - check results carefully
```

DHARMa residual



##residual plots show that there is overdispersion and excessive outliers. Plot negative binomial

Plot negative binomial to see if that fixes the overdispersion.

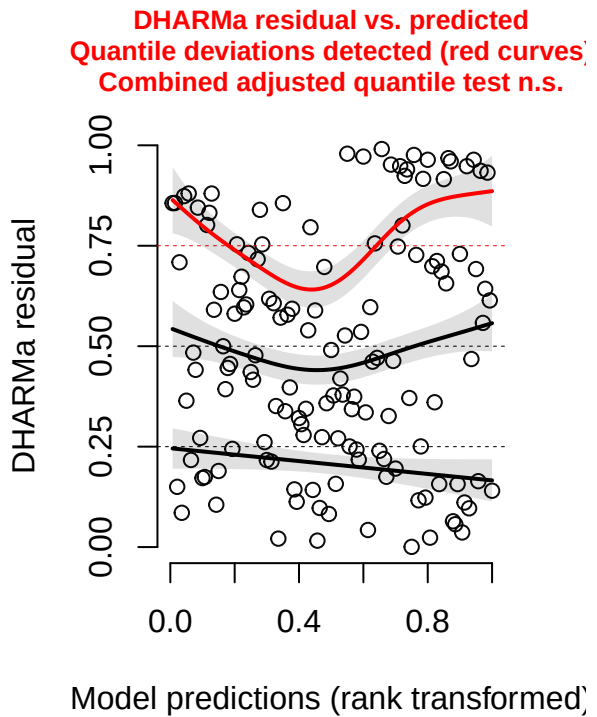
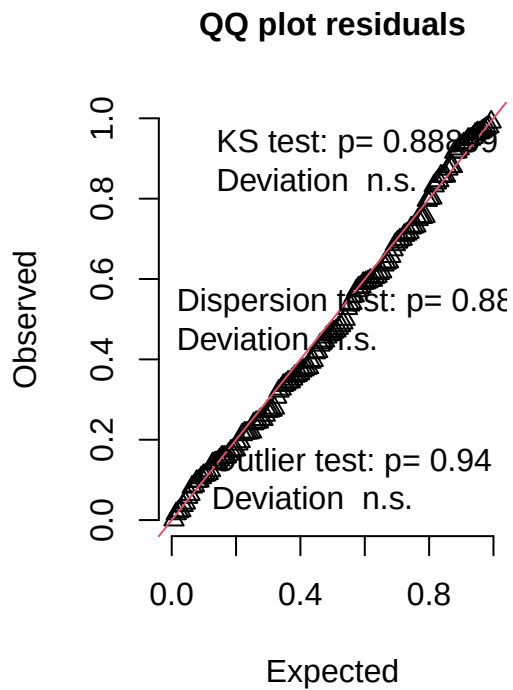
```
burrows_lm_negbin <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
  julian.date_scaled,
  family = nbinom1,
  data = field_data)
```

Diagnostics

#simulate residuals

```
residuals_burrows_negbin <- simulateResiduals(fittedModel = burrows_lm_negbin, plot = TRUE)
```

DHARMa residual

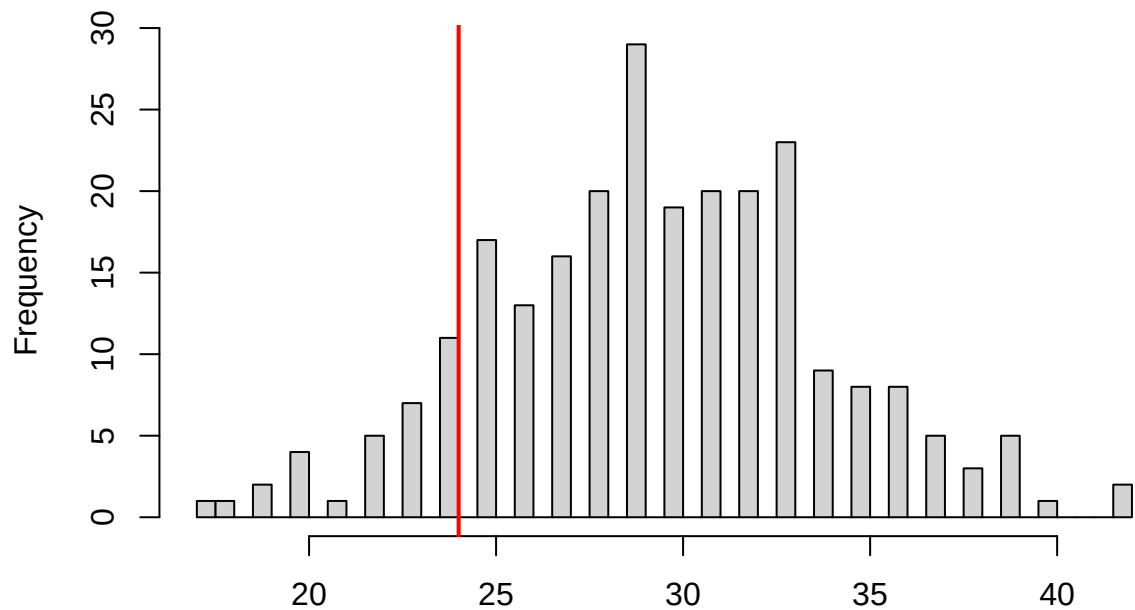


##residual plots show that there is no overdispersion or excessive outliers. Now check zero-inflation.

Test zero-inflation

```
testZeroInflation(residuals_burrows_negbin)
```

DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.256

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.81235, p-value = 0.256
## alternative hypothesis: two.sided
```

##p-value = 0.256. No strong evidence of zero-inflation. I'll try the zero-inflated model to see if it

Plot zero-inflated negative binomial and recheck diagnostics.

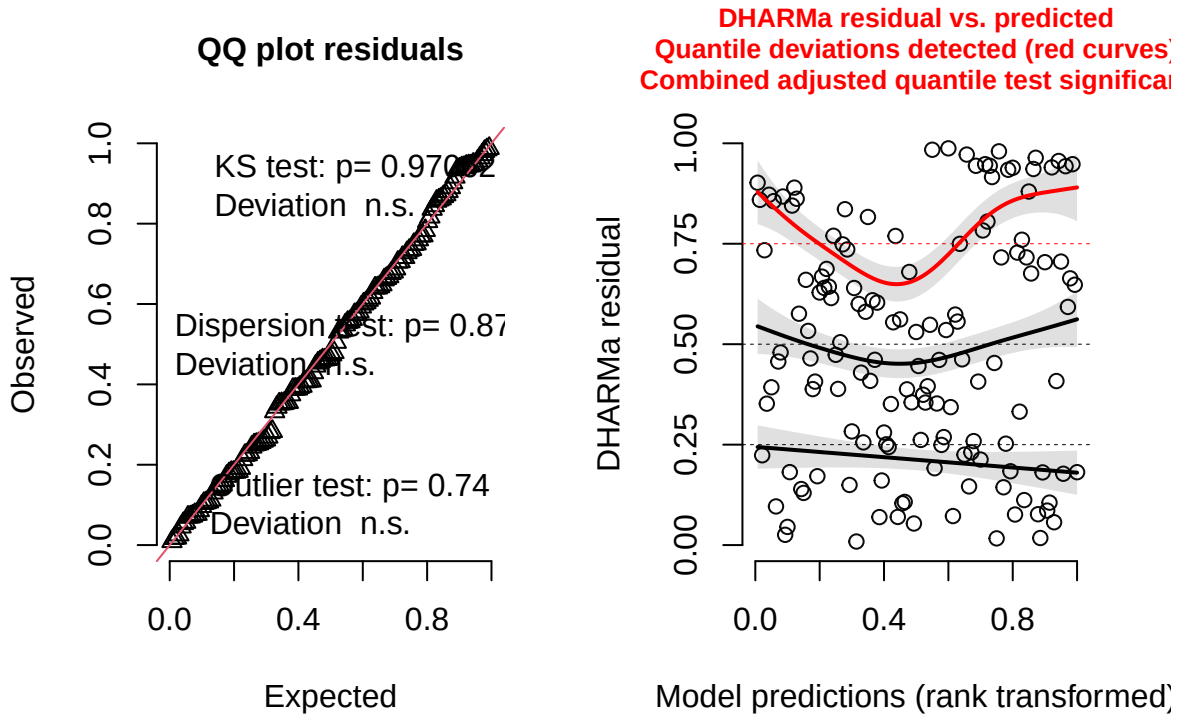
```
burrows_lm_zero_negbin <- glmmTMB(num_burrows ~ grass_forb_perc_scaled +
  julian.date_scaled,
  ziformula = ~1, # zero-inflation model
  family = nbinom1,
  data = field_data)
```

Diagnostics

```
#simulate residuals
```

```
residuals_burrows_zero_negbin <- simulateResiduals(fittedModel = burrows_lm_zero_negbin, plot = TRUE)
```

DHARMA residual



##residual plots does not look significantly better than the non-zero-inflated model.

Compare AIC values between the 2 models

```
summary(burrows_lm_negbin)
```

```
## Family: nbinom1 ( log )
## Formula: num_burrows ~ grass_forb_perc_scaled + julian.date_scaled
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    1052.0    1063.7    -522.0    1044.0      136
##
##
## Dispersion parameter for nbinom1 family (): 48.9
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    3.11709    0.12941  24.086 < 2e-16 ***
## grass_forb_perc_scaled 0.43025    0.10936   3.934 8.35e-05 ***
## julian.date_scaled    0.08301    0.08634   0.961  0.336
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(burrows_lm_zero_negbin)
```

```
## Family: nbinom1 ( log )
## Formula:      num_burrows ~ grass_forb_perc_scaled + julian.date_scaled
## Zero inflation: ~1
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    1054.0    1068.7    -522.0    1044.0      135
##
##
## Dispersion parameter for nbinom1 family (): 48.9
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.11709    0.12941  24.086 < 2e-16 ***
## grass_forb_perc_scaled 0.43026    0.10936   3.934 8.35e-05 ***
## julian.date_scaled    0.08301    0.08634   0.961  0.336
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    -20.08    4667.00  -0.004   0.997
```

##AIC and BIC values are slightly lower for negative binomial and there was no strong evidence for zero

Results for the final model

```
summary(burrows_lm_negbin)
```

```
## Family: nbinom1 ( log )
## Formula:      num_burrows ~ grass_forb_perc_scaled + julian.date_scaled
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    1052.0    1063.7    -522.0    1044.0      136
##
##
## Dispersion parameter for nbinom1 family (): 48.9
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.11709    0.12941  24.086 < 2e-16 ***
## grass_forb_perc_scaled 0.43025    0.10936   3.934 8.35e-05 ***
## julian.date_scaled    0.08301    0.08634   0.961  0.336
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

GLM #4: Number of inactive rodent burrows

First, plot poisson

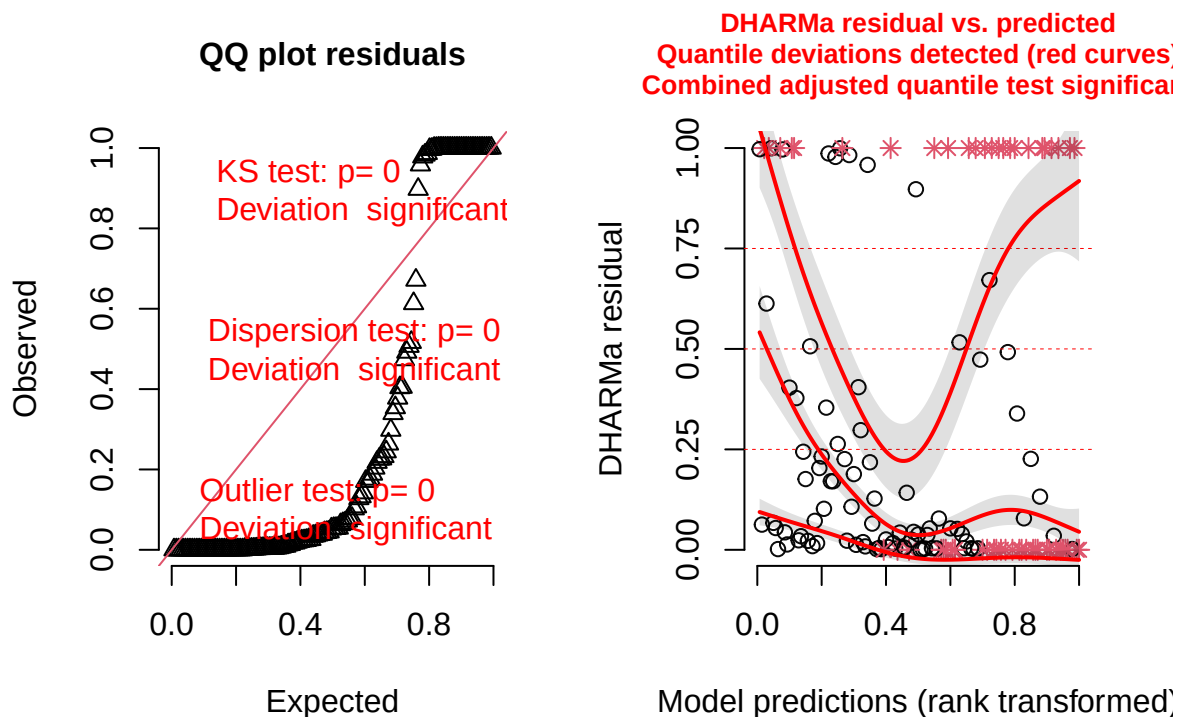
```
inactiveburrows_lm_pois <- glmmTMB(num_inactive_burrows ~ grass_forb_perc_scaled +
  julian.date_scaled,
  family = poisson, #poisson
  data = field_data)
```

Diagnostics

```
#simulate residuals
```

```
residuals_inactiveburrows_pois <- simulateResiduals(fittedModel = inactiveburrows_lm_pois, plot = TRUE)
```

DHARMA residual



```
##residual plots show that there is overdispersion and excessive outliers. Plot negative binomial
```

Plot negative binomial to see if that fixes the overdispersion.

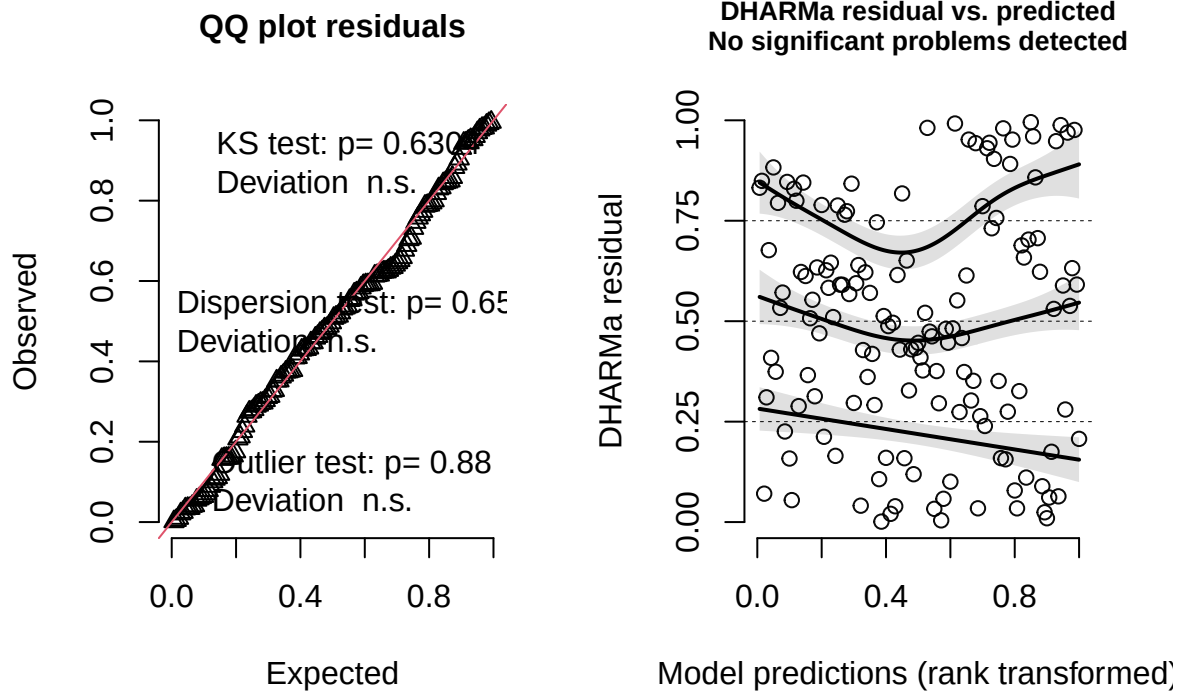
```
inactiveburrows_lm_negbin <- glmmTMB(num_inactive_burrows ~ grass_forb_perc_scaled +
  julian.date_scaled,
  family = nbinom1,
  data = field_data)
```

Diagnostics

```
#simulate residuals
```

```
residuals_inactiveburrows_negbin <- simulateResiduals(fittedModel = inactiveburrows_lm_negbin, plot = T)
```

DHARMA residual

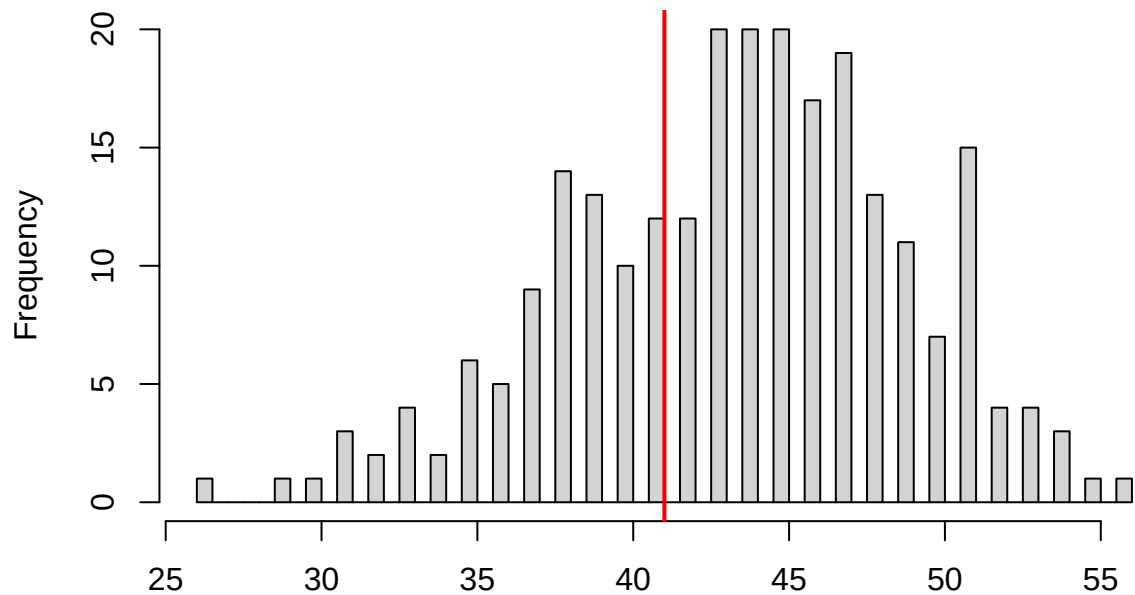


##residual plots show that there is no overdispersion or excessive outliers. Now check zero-inflation.

Test zero-inflation

```
testZeroInflation(residuals_inactiveburrows_negbin)
```


DHARMA zero-inflation test via comparison to expected zeros with simulation under H0 = fitted model



Simulated values, red line = fitted model. p-value (two.sided) = 0.664

```
##
## DHARMA zero-inflation test via comparison to expected zeros with
## simulation under H0 = fitted model
##
## data: simulationOutput
## ratioObsSim = 0.94158, p-value = 0.664
## alternative hypothesis: two.sided
```

##p-value = 0.664. No strong evidence of zero-inflation. I'll try the zero-inflated model to see if it

Plot zero-inflated negative binomial and recheck diagnostics.

```
inactiveburrows_lm_zero_negbin <- glmmTMB(num_inactive_burrows ~ grass_forb_perc_scaled +
  julian.date_scaled,
  ziformula = ~1, # zero-inflation model
  family = nbinom1,
  data = field_data)
```

Compare AIC values between the 2 models

```
summary(inactiveburrows_lm_negbin)
```

```
## Family: nbinom1 ( log )
## Formula:
```

```
## num_inactive_burrows ~ grass_forb_perc_scaled + julian.date_scaled
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    817.7    829.4   -404.8    809.7      136
##
##
## Dispersion parameter for nbinom1 family (): 22
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.20210    0.13894  15.850 < 2e-16 ***
## grass_forb_perc_scaled 0.36568    0.12113   3.019  0.00254 **
## julian.date_scaled    0.08254    0.09534   0.866  0.38663
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(inactiveburrows_lm_zero_negbin)
```

```
## Family: nbinom1 ( log )
## Formula:
## num_inactive_burrows ~ grass_forb_perc_scaled + julian.date_scaled
## Zero inflation: ~1
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    819.7    834.4   -404.8    809.7      135
##
##
## Dispersion parameter for nbinom1 family (): 22
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.20210    0.13894  15.849 < 2e-16 ***
## grass_forb_perc_scaled 0.36567    0.12113   3.019  0.00254 **
## julian.date_scaled    0.08253    0.09534   0.866  0.38667
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -18.04    5337.31  -0.003   0.997
```

##AIC and BIC values are slightly lower for negative binomial and there was no strong evidence for zero

Results for the final model

```
summary(inactiveburrows_lm_negbin)
```

```
## Family: nbinom1 ( log )
## Formula:
## num_inactive_burrows ~ grass_forb_perc_scaled + julian.date_scaled
```

```
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    817.7    829.4   -404.8    809.7      136
##
##
## Dispersion parameter for nbinom1 family (): 22
##
## Conditional model:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    2.20210    0.13894  15.850 < 2e-16 ***
## grass_forb_perc_scaled 0.36568    0.12113   3.019  0.00254 **
## julian.date_scaled    0.08254    0.09534   0.866  0.38663
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

GLM #5: Floral abundance

Plot gamma regression since distribution of floral abundance across sites is positive and right-skewed.

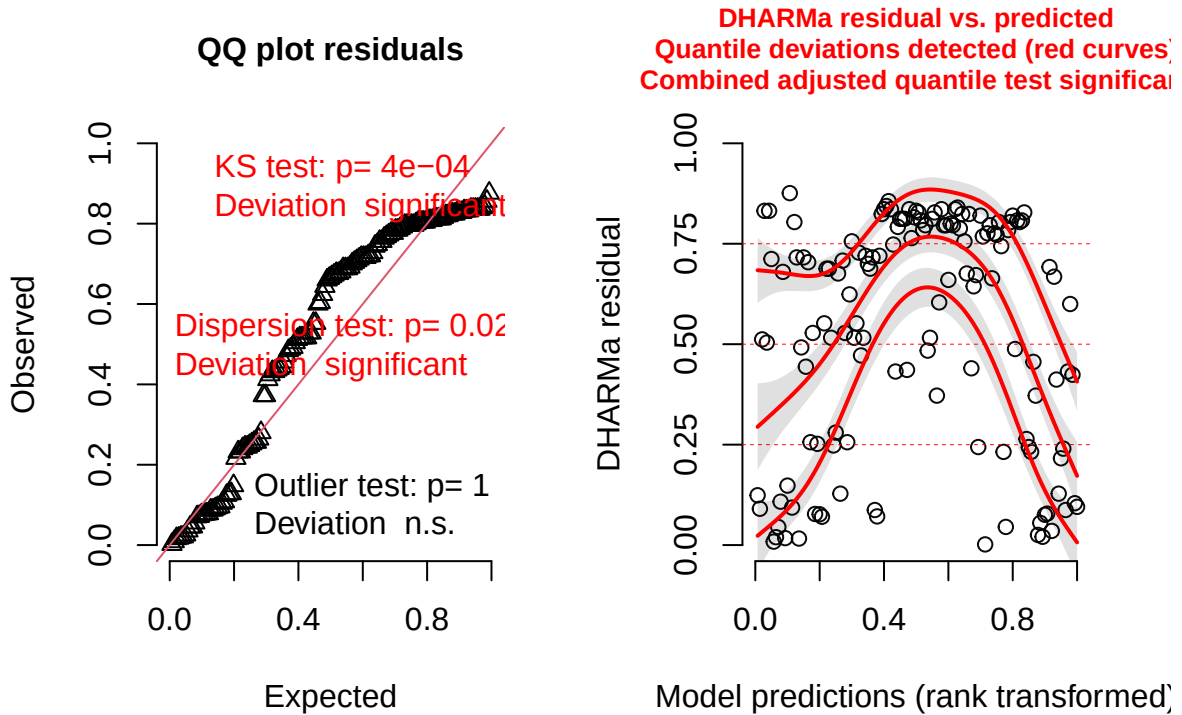
```
floral_lm_tweedie <- glmmTMB(bombus_floral_abun ~ grass_forb_perc_scaled +
  julian.date_scaled,
  family = tweedie(link = "log"), ##use tweedie family instead of Gamma since it can ha
  data = field_data)
```

Diagnostics

```
#simulate residuals

residuals_floral_tweedie <- simulateResiduals(fittedModel = floral_lm_tweedie, plot = TRUE)
```

DHARMA residual



##residual plots show that there is overdispersion and excessive outliers.

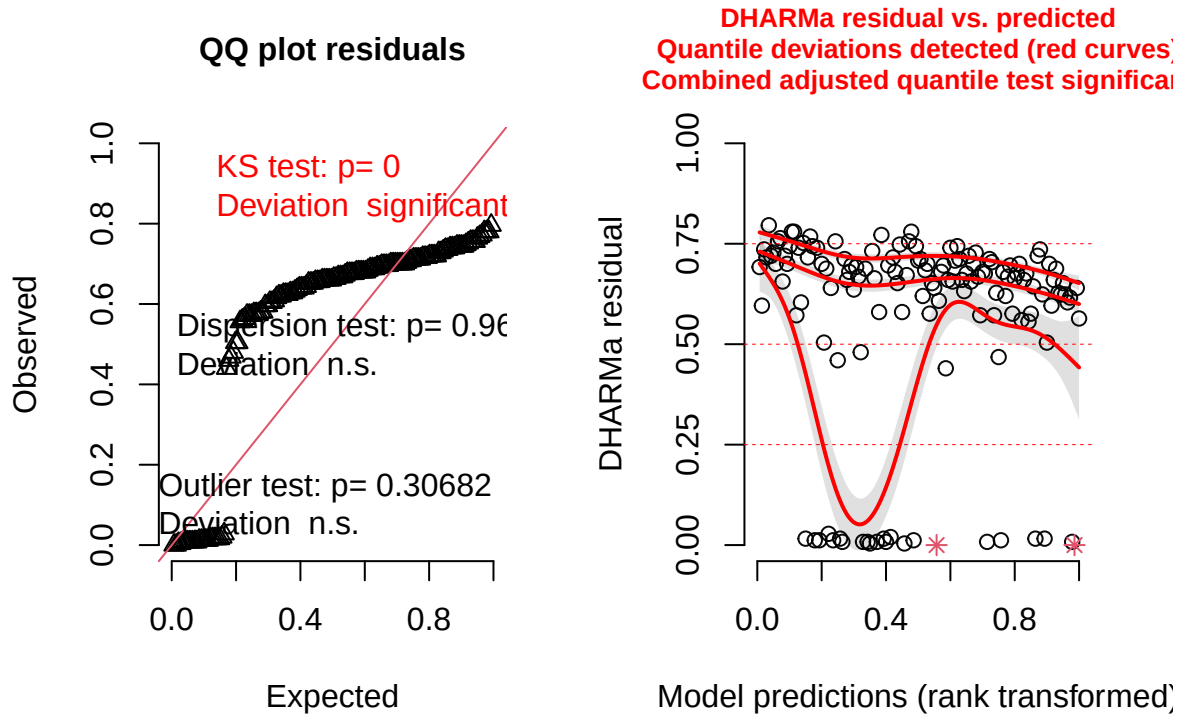
Try plotting the $\log(\text{floral abundance})$ as a linear regression

```
field_data$log_bombus_floral_abun <- log(field_data$bombus_floral_abun + 1e-4)
floral_lm_log <- lm(log_bombus_floral_abun ~ grass_forb_perc_scaled + julian.date_scaled, data = field_data)
```

Diagnostics

```
residuals_floral_log <- simulateResiduals(fittedModel = floral_lm_log, plot = TRUE)
```

DHARMA residual



##fixed overdispersion and outliers. KS test suggests non-normality but I will check this in the Q-Q plot

```
library(car)
```

```
## Loading required package: carData
```

```
##
```

```
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
```

```
##
```

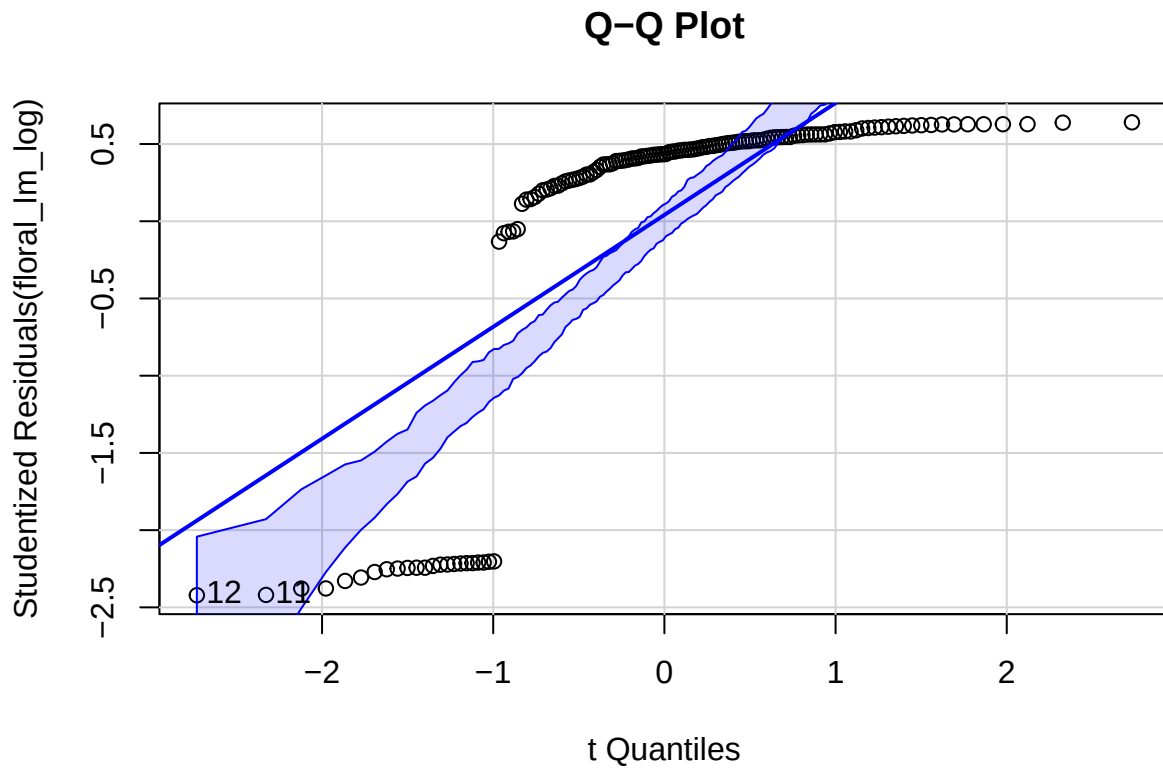
```
## recode
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
## some
```

```
qqPlot(floral_lm_log, main = "Q-Q Plot")
```



```
## [1] 11 12
```

```
##Q-Q plot shows that residuals are still not normal. Need to try a different model.
```

Try running Tweedie model again but adjusting for overdispersion by adding a random effect for data structure

```
floral_glmm_tweedie <- glmmTMB(bombus_floral_abun ~ grass_forb_perc_scaled +
  julian.date_scaled + (1|area/site_id/plot_id),
  family = tweedie(link = "log"),
  data = field_data)
```

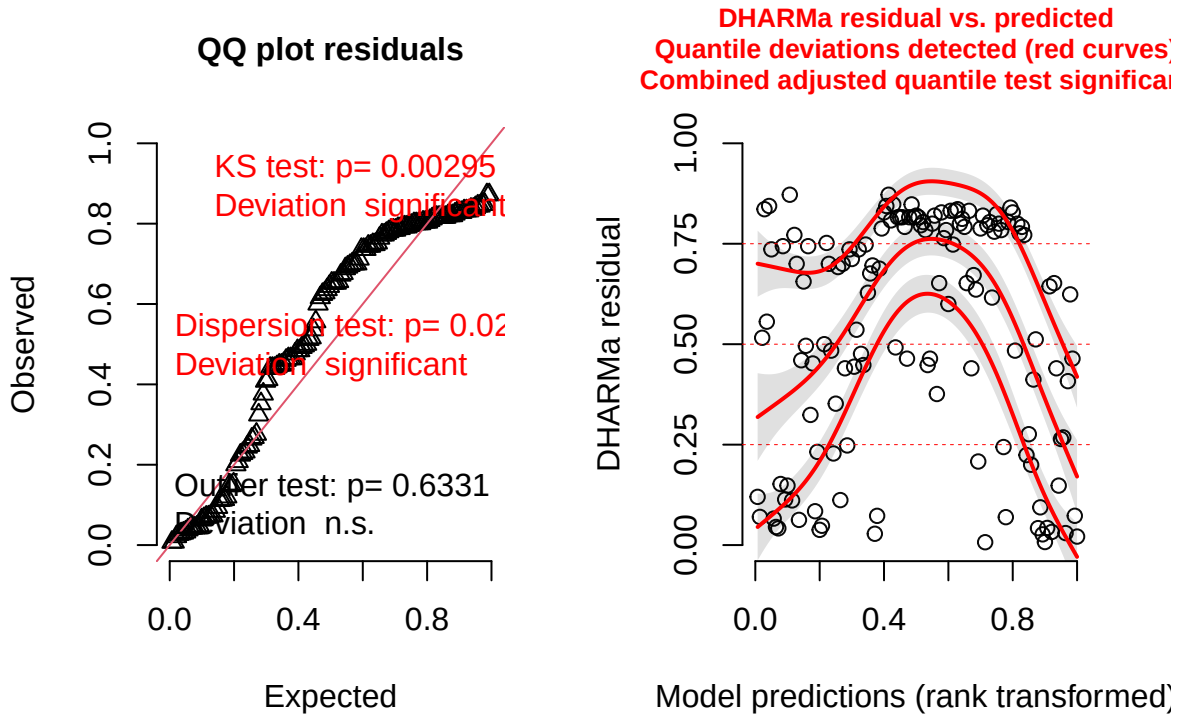
Diagnostics

```
#simulate residuals
```

```
residuals_floral_tweedie_mixed <- simulateResiduals(fittedModel = floral_glmm_tweedie, plot = TRUE)
```

```
## Warning in newton(lsp = lsp, X = G$X, y = G$y, Eb = G$Eb, UrS = G$UrS, L = G$L,
## : Fitting terminated with step failure - check results carefully
```

DHARMa residual



##residual plots show that there is still overdispersion. try different random effect.

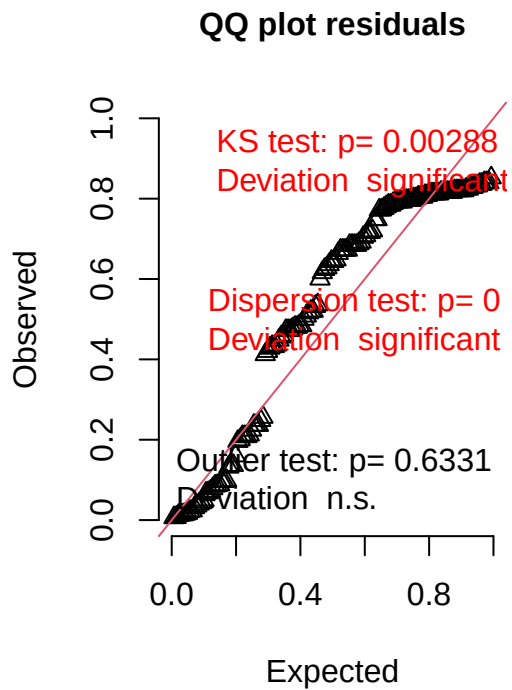
Try simpler random effect structure (ignore site)

```
floral_glmm_tweedie2 <- glmmTMB(bombus_floral_abun ~ grass_forb_perc_scaled +
  julian.date_scaled + (1|area/plot_id),
  family = tweedie(link = "log"),
  data = field_data)
```

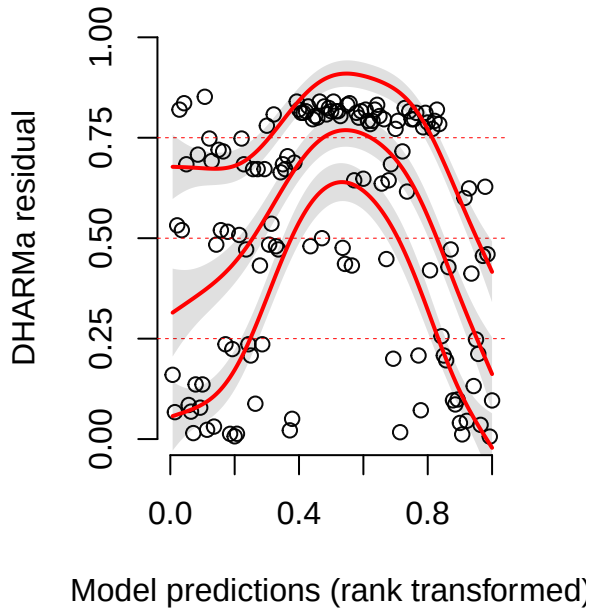
Diagnostics –check for overdispersion

```
sim_res <- simulateResiduals(floral_glmm_tweedie2)
plot(sim_res)
```

DHARMa residual

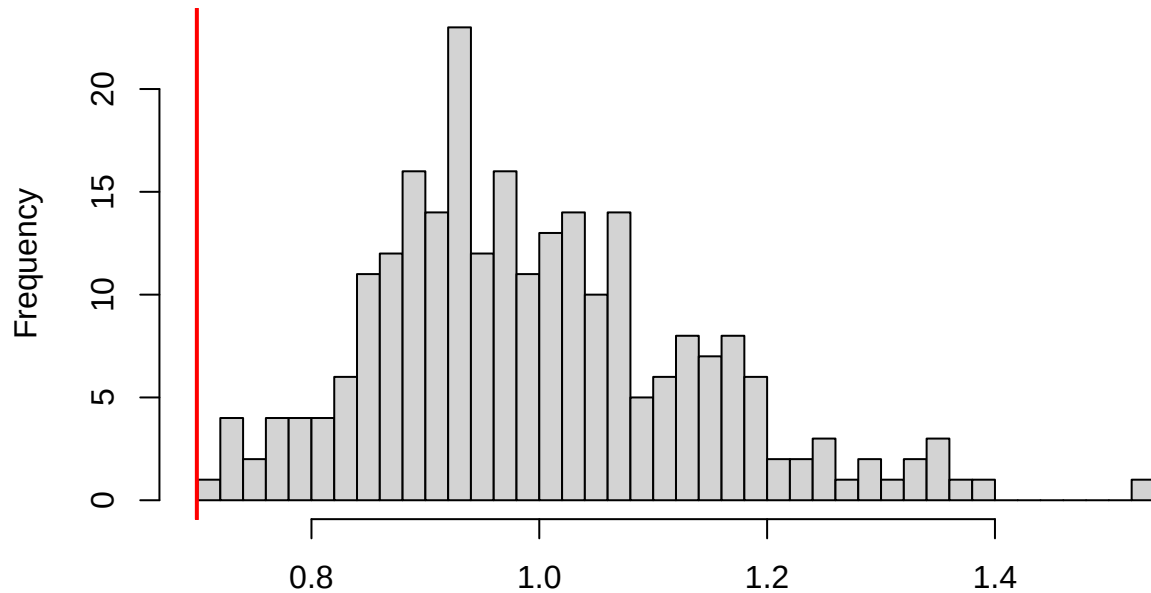


DHARMa residual vs. predicted
Quantile deviations detected (red curves)
Combined adjusted quantile test significant



```
testDispersion(sim_res)
```


DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated



Simulated values, red line = fitted model. p-value (two.sided) = 0

```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.70199, p-value < 2.2e-16
## alternative hypothesis: two.sided
```

##still overdispersion. Keep original random effect structure for now but try modelling zero-inflation

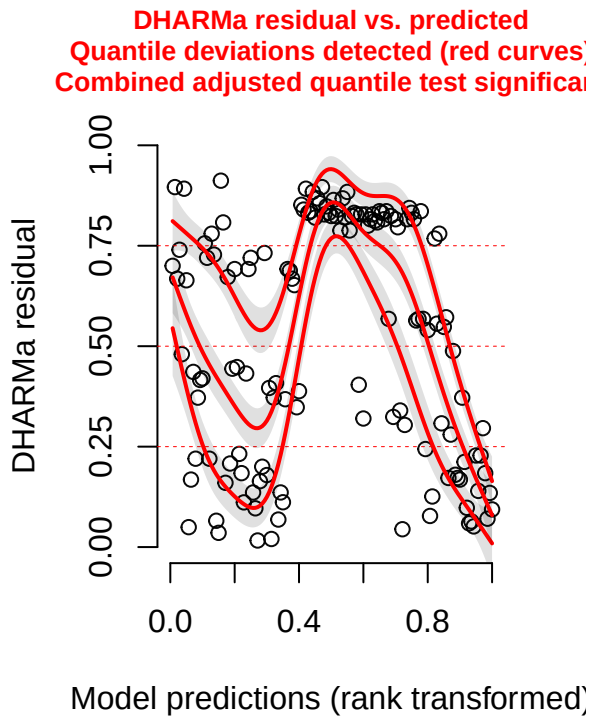
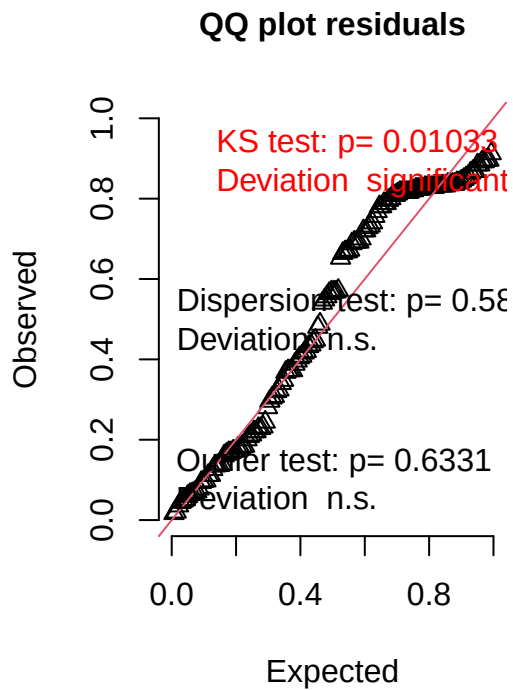
Zero-inflated model

```
floral_glmm_tweediezero <- glmmTMB(bombus_floral_abun ~ grass_forb_perc_scaled +
  julian.date_scaled + (1|area/site_id/plot_id),
  ziformula = ~1, #zero-inflated model
  family = tweedie(link = "log"),
  data = field_data)
```

Diagnostics –check for overdispersion

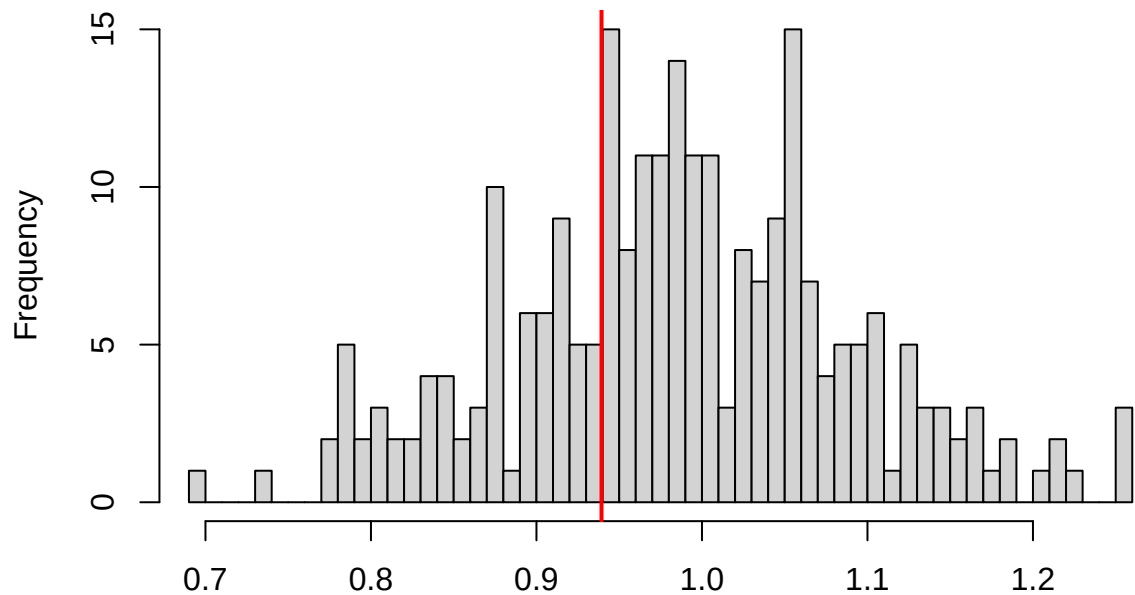
```
sim_res <- simulateResiduals(floral_glmm_tweediezero)
plot(sim_res)
```

DHARMa residual



```
testDispersion(sim_res)
```

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated



Simulated values, red line = fitted model. p-value (two.sided) = 0.584

```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.95183, p-value = 0.584
## alternative hypothesis: two.sided
```

##no more overdispersion!

Check other random effect structures to determine which is best

```
#plot nested in area (drop site)

floral_glmm_tweediezero_plot <- glmmTMB(bombus_floral_abun ~ grass_forb_perc_scaled +
  julian.date_scaled + (1|area/plot_id),
  ziformula = ~1, #zero-inflated model
  family = tweedie(link = "log"),
  data = field_data)

#site nested in area (drop plot)

floral_glmm_tweediezero_site <- glmmTMB(bombus_floral_abun ~ grass_forb_perc_scaled +
  julian.date_scaled + (1|area/site_id),
  ziformula = ~1, #zero-inflated model)
```

```

        family = tweedie(link = "log"),
        data = field_data)

#plot only

floral_glmm_tweediezero_plotonly <- glmmTMB(bombus_floral_abun ~ grass_forb_perc_scaled +
      julian.date_scaled + (1|plot_id),
      ziformula = ~1, #zero-inflated model
      family = tweedie(link = "log"),
      data = field_data)

#site only

floral_glmm_tweediezero_siteonly <- glmmTMB(bombus_floral_abun ~ grass_forb_perc_scaled +
      julian.date_scaled + (1|site_id),
      ziformula = ~1, #zero-inflated model
      family = tweedie(link = "log"),
      data = field_data)

#area only

floral_glmm_tweediezero_areaonly <- glmmTMB(bombus_floral_abun ~ grass_forb_perc_scaled +
      julian.date_scaled + (1|area),
      ziformula = ~1, #zero-inflated model
      family = tweedie(link = "log"),
      data = field_data)

#check AIC
AIC(
  floral_glmm_tweediezero,
  floral_glmm_tweediezero_plot,
  floral_glmm_tweediezero_site,
  floral_glmm_tweediezero_plotonly,
  floral_glmm_tweediezero_siteonly,
  floral_glmm_tweediezero_areaonly
)

```

```

##              df      AIC
## floral_glmm_tweediezero      9 516.2534
## floral_glmm_tweediezero_plot    8 514.2534
## floral_glmm_tweediezero_site    8 513.1628
## floral_glmm_tweediezero_plotonly 7 487.1094
## floral_glmm_tweediezero_siteonly 7 512.3448
## floral_glmm_tweediezero_areaonly 7 512.2534

```

##plot only model has lowest AIC--> go with that one!

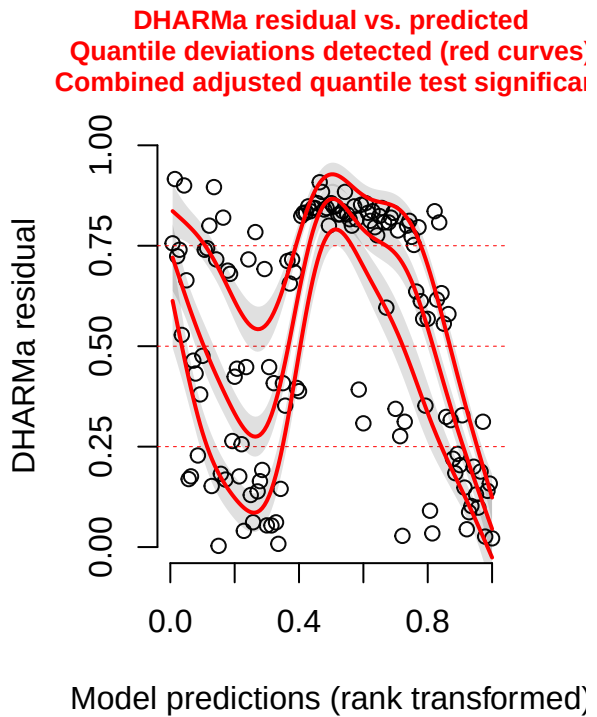
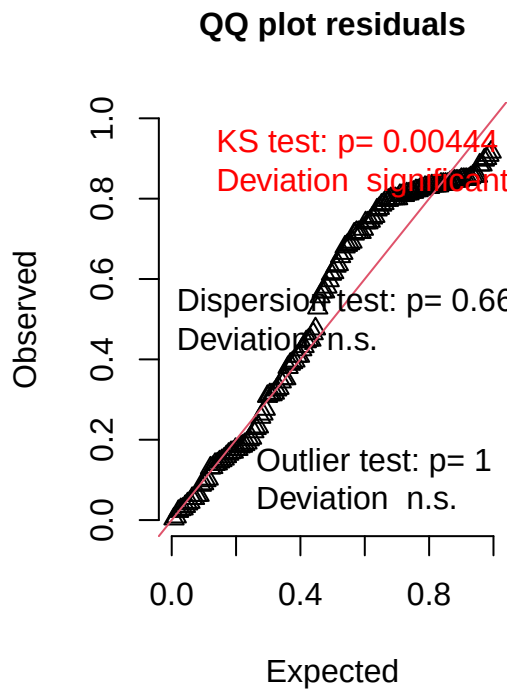
Check overdispersion in plot only model

```

sim_res <- simulateResiduals(floral_glmm_tweediezero_plotonly)
plot(sim_res)

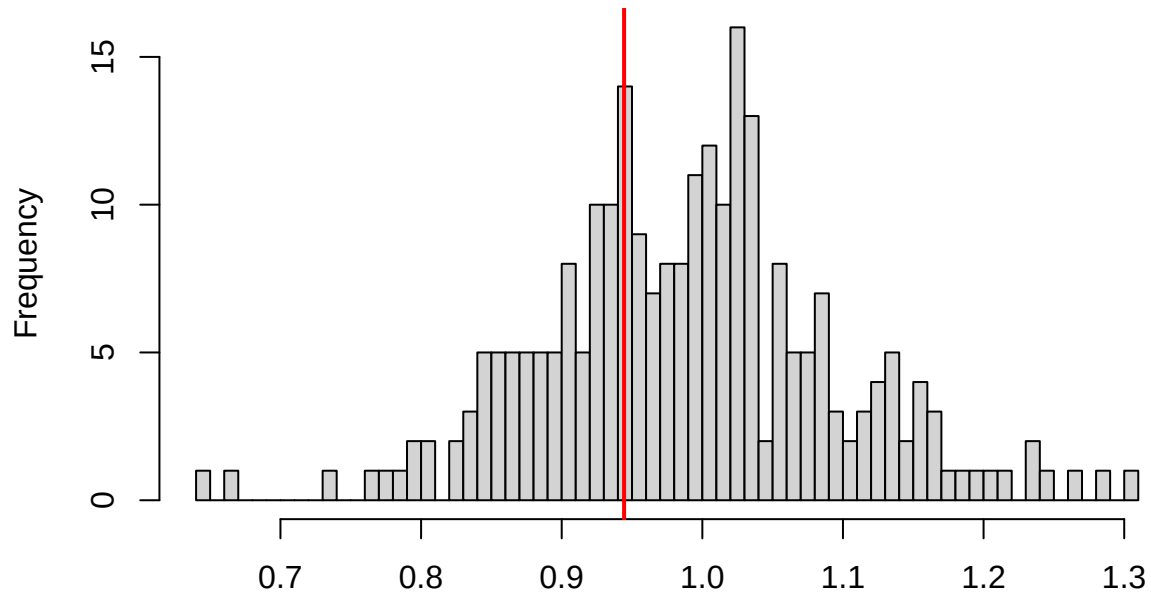
```

DHARMa residual



```
testDispersion(sim_res)
```

DHARMA nonparametric dispersion test via sd of residuals fitted vs. simulated



Simulated values, red line = fitted model. p-value (two.sided) = 0.664

```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.95435, p-value = 0.664
## alternative hypothesis: two.sided
```

##still no overdispersion

Look at summary for final model

```
summary(floral_glmm_tweediezero_plotonly)
```

```
## Family: tweedie ( log )
## Formula:
## bombus_floral_abun ~ grass_forb_perc_scaled + julian.date_scaled +
## (1 | plot_id)
## Zero inflation: ~1
## Data: field_data
##
##      AIC      BIC    logLik -2*log(L)  df.resid
##    487.1    507.7   -236.6    473.1     133
##
```

```

## Random effects:
##
## Conditional model:
##   Groups   Name      Variance Std.Dev.
## plot_id (Intercept) 1.087e-10 1.043e-05
## Number of obs: 140, groups: plot_id, 2
##
## Dispersion parameter for tweedie family (): 0.5
##
## Conditional model:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.068626   0.040817  26.181   <2e-16 ***
## grass_forb_perc_scaled -0.009948   0.047577  -0.209    0.834
## julian.date_scaled    -0.067595   0.041604  -1.625    0.104
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Zero-inflation model:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.6457    0.2318   -7.1 1.25e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```